COMPUTER SCIENCE

Course Overview

The IB Diploma **Programme** Computer Science course explores the fundamentals of computation through a blend of theoretical foundations and practical programming skills. It **emphasises** the development of problem-solving abilities using **abstraction, algorithmic thinking, and rigorous analysis**. Students engage with core topics such as programming, computer systems, networks, databases, and emerging technologies.

Learning Aims

The aims of computer science course for SL and HL pupils are to:

1. The aims of the course are to enable students to:
2. Develop conceptual understanding across the subject and other DP disciplines.
3. Acquire and apply a body of knowledge, methods, and tools characterising computer science.
4. Analyse and evaluate solutions developed through computational thinking.
5. Approach unfamiliar situations with creativity and resilience.
6. Design and implement solutions to local and global problems.
7. Appreciate the possibilities and limitations of the field.
8. Evaluate the impact of emerging technologies.
9. Communicate and collaborate effectively.
10. Develop awareness of the environmental, economic, and social impacts of computer science.

Course Objectives

There are four assessment objectives for Computer Science. Having followed this course at SL or HL, Students are expected to demonstrate:

- **AO1:** Knowledge and understanding of facts, concepts, and principles.
- **AO2:** Application and use of computational thinking strategies and design methods.
- **AO3:** Construction, synthesis, **analysis**, and evaluation of system models and ethical implications.
- **AO4:** Technical proficiency in **coding and real-world problem solving**.

.

Unit Overview

TIMEFRAME	Unit Title	Curriculum Mapping	Topics
Year 1			
1–3	**Unit 1: Program Elements**	B2.1.1–B2.1.5	Variables, data types, input/output, string ops
4–7	**Unit 2: Control Structures**	B2.3.1–B2.3.3	Selection (if/else), Iteration (for/while)
8–10	**Unit 3: Modularisation**	B2.3.4	Functions, scope, and tracing with dry runs
11–13	**Unit 4: Computational Thinking**	B2.4.1–B2.4.3	Searching/sorting algorithms (linear/binary, bubble)
14–15	**Unit 5: Abstraction**	B1.1.1–B1.1.2	Problem specification, decomposition, and abstraction
16–17	**Unit 6: Algorithm Design**	B1.1.4, B2.4	Flowcharts, pseudocode, and trace tables
17	**IA Checkpoint #1**	Internal Assessment	**IA Proposal Submission**
18–19	**Unit 7: Architecture**	A1.1.1–A1.1.5	CPU components, memory (RAM/ROM), FDE cycle
20–21	**Unit 8: Logic & OS**	A1.2.3–A1.3.4	Logic gates, truth tables, OS scheduling, polling
22–24	**Unit 9: Database Concepts**	A3.1.1–A3.2.3	Relational features, ERDs, and data types
25–27	**Unit 10: SQL & Normalisation**	A3.2.5–A3.3.3	1NF–3NF, SQL CRUD operations (Joins, Updates)
28	**IA Checkpoint #2**	Internal Assessment	**IA Progress Check**

	Unit	Curriculum Mapping	Lesson Topics/Content	IA Milestone
1–2	Unit 11: OOP Fundamentals (Pt 1)	B3.1.1–B3.1.5	Classes, Objects, Attributes, and Encapsulation.	IA Criterion D: Commencement of product development.
3	Unit 11: OOP Fundamentals (Pt 1)	B3.1.1–B3.1.5	Continued: Static vs non-static members and object modelling.	
4	Unit 12: OOP Fundamentals (Pt 2)	B3.1.1–B3.1.5	Methods, Constructors, and Modular classes.	
5	Unit 12: OOP Fundamentals (Pt 2)	B3.1.1–B3.1.5	Designing class behaviour and modularity.	
6	Unit 13: Advanced OOP & Recursion	B3.2.1–B3.2.5	Inheritance, Polymorphism, and Design patterns.	
7	Unit 13: Advanced OOP & Recursion	B2.4.4–B2.4.5	Recursion fundamentals (HL only).	IA Progress Check: Alpha version demonstrated to the client.
8	Unit 14: Database Extension (HL)	A3.3.4–A3.3.6	Aggregate functions, Database views, and ACID transactions (HL only).	
9	Unit 14: Database Extension (HL)	A3.4	NoSQL and Data warehouses (HL only).	
10	Unit 15: ADTs & Hash Tables (HL)	B4.1.1–B4.1.2	Stacks and Queues memory usage; Sets (HL only).	
11	Unit 15: ADTs & Hash Tables (HL)	B4.1.5–B4.1.6	Hash tables (HL only).	
12	Unit 16: Linked Lists & Trees (HL)	B4.1.3–B4.1.4	Singly and Doubly linked lists (HL only).	
13	Unit 16: Linked Lists & Trees (HL)	B4.1.3–B4.1.4	Binary Search Trees (BSTs) (HL only).	
14–15	Unit 17: Networking & Security	A2.1.1–A2.1.5	LAN/WAN/VPN, Protocols, and TCP/IP Model.	IA Criterion D: Completion of core functionality and code comments.
16	Unit 17: Networking & Security	A2.2.2, A2.4	Servers, Firewalls, and Encryption.	
17	Unit 17: Networking & Security	A2.4	Vulnerabilities and Countermeasures (HL only).	
18	Unit 18: Transmission	A2.2.1, A2.2.3	Client-server vs P2P and Packet switching.	
19	Unit 18: Transmission	A2.3.1–A2.3.4	IP addressing (IPv4/v6) and Routing logic (HL only).	
20	Unit 19: Advanced Fundamentals	A1.1.8–A1.1.9	Compression, Cloud services, and Multitasking (HL only).	

21	Unit 19: Advanced Fundamentals	A1.3.5–A1.4	Control systems and Translation processes (HL only).	
22	Unit 20: Data Structures & Files	B2.2	Static vs Dynamic structures, Arrays and Lists.	IA Criterion E: Formal evaluation and client feedback.
23	Unit 20: Data Structures & Files	B2.5	Stacks, Queues, and File processing (read/write/append).	
24	Unit 21: Advanced Machine Learning	A4.1.1–A4.2.2	ML types, Preprocessing, and Association rules (HL only).	
25	Unit 21: Advanced Machine Learning	A4.3.1–A4.3.10	Reinforcement learning, Genetic algorithms, and ANN/CNN (HL only).	
26	Unit 22: Case Study	N/A	Model selection (HL only). Case Study Introduction.	
27	Unit 23: Exam Review	N/A	Emerging Technologies: Scenario research and Paper 1 Section B prep.	

Assessment

While the class will use formative assessment and reporting strategies employed at the school, final assessment in fulfilment of course requirements counting against the IB Diploma will be based on the following breakdown of requirements: (first assessment – 2027)

IB Component Breakdown

Assessment Outline – SL	
Assessment Component	**Weighting**
External Assessment (2 hours 30 minutes) **Paper 1: (1 hour 15 minutes)** Section A extended response questions linked to theme A: Concepts of computer science Section B short-response questions linked to the pre-seen case study (50 marks) **Paper 2: (1 hour and 15 minutes)** Extended response questions linked to theme B: Computational thinking and problem-solving (50 marks)	**70%** 35% 35%
Internal assessment (35 hours) This component is internally assessed by the teacher and externally moderated by the IB at the end of the course. IA consists of one task: the computational solution (30 marks)	**30%**

ASSESSMENT OUTLINE – HL	
Assessment Component	**Weighting**
External Assessment (4 hours) **Paper 1: (2 hours)** Section A extended response questions linked to theme A: Concepts of computer science Section B short-response questions linked to the pre-seen case study (80 marks) **Paper 2: Database Theory (2 hours)** Extended-response questions linked to theme B: Computational thinking and problem-solving. (80 marks)	**80%** 40% 40%
Internal assessment This component is internally assessed by the teacher and externally moderated by the IB at the end of the course. IA consists of one task: the computational solution (30 marks)	**20%**

The external assessment of the Diploma Programme Computer Science course consists of two examination papers at SL and three examination papers at HL that are externally set and externally marked. They are designed to allow pupils to demonstrate their competencies in relation to the business management assessment objectives.

External Assessment Details

Paper 1 (SL | HL)

- **Duration:** 1 hour 15 minutes | 2 hours
- **Weighting:** 35% | 40%
- **Marks:** 50 for SL | 80 for HL (All questions compulsory)
- **Format:** Completed together without interruptions.
    - **Section A:** Extended-response questions based on SL Theme A: Concepts of Computer Science.
    - **Section B:** Short-response questions based on the pre-seen case study and its two challenge questions.

Case Study Overview

- **Purpose:** Stimulus to research current developments, emerging technologies, or ethical issues. The SL case study features a scenario with two challenge questions to guide research.
- **Release:** Published on the Programme Resource Centre 12 months before the May exam
- **Research Methods:** Primary/secondary sources, interviews, guest speakers, and field trips.

Assessment Objectives (AOs) Tested:

- **Show** understanding of system workings and computational thinking fundamentals.
- **Apply** syllabus concepts to the case study context.
- **Explain** connections between case study scenarios and other scenarios.
- **Discuss** relevant impacts and ethical issues.
- **Evaluate, formulate, and justify** strategies using case study data, personal research, and new exam stimulus.

Paper 2 (SL | HL)

- **Duration:** 1 hour 15 minutes | 2 hours
- **Weighting:** 35% | 40%
- **Marks:** 50 for SL | 80 for HL (All questions compulsory)

Paper 2 consists of extended-response questions linked to SL topics in theme B: Computational thinking and problem-solving.

One of the questions will focus on algorithmic thinking without the need to interpret or write code.

The Python programming language has a number of built-in functions that can help with finding solutions to common computational challenges. The use of functions such as ***sort, pop, len, max and min*** can be very useful when programming algorithms and their use is encouraged in tasks such as the IA (the computational solution). However, in order to assess the full range of students' abilities, certain questions in the examination will prohibit specific built-in functions. These instances will be clearly indicated in the examination papers.

Internal Assessment

Assessment Description:

Internal assessment is an integral part of the course and is compulsory for both SL and HL students. It enables students to demonstrate the application of their skills and knowledge, and to pursue their personal interests, without the time limitations and other constraints that are associated with written examinations. Students are required to produce a solution that consists of a cover page, the product and the documentation. The focus of the solution is on providing either an original product or additional functionality to an existing product for a client.

Solution Development:

The requirement of the internal assessment is to develop a solution for a specified client to a specified problem or an unanswered question.
The solution is assessed using five criteria.

A. Problem Specification
B. Planning
C. System Overview
D. Development
E. Evaluation

Learning Environment

In line with IB pedagogy, this course is **inquiry-based and student-centred**. Students will learn to program using Python, while theories of computer fundamentals and networking are integrated to support system design. The Internal Assessment is the hallmark of the course, requiring students to propose and manage change for a real-world client.

Course design: August 2026 by Rhoxette Nazi P. Beatingo